

Faisal Hossain Raquib

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[Personal Website](#) | [Google Scholar](#) | [Faisal](#) | [Faisal](#)

Education

Rajshahi University of Engineering & Technology (RUET)

B.Sc. in Electrical & Computer Engineering

Rajshahi, Bangladesh

February 2019 – December 2024

• CGPA: 3.18/4.00 | Final 4 semesters: 3.56/4.00

Research Interests

- Machine Learning
- Deep Learning
- Natural Language Processing
- Large Language Models
- Low-Resource Language Processing
- Multimodal Learning
- Explainable AI
- Trustworthy AI
- Generative AI

Publications

BanHADEX: Towards Explainable HATE Speech Detection in Bangla Using Human Annotated EXplanation 2026

Faisal Hossain Raquib, Akm Moshir Rahman Mazumder, Md Fahim, Md Tahmid Hasan Fuad, Md Farhan Ishmam, Faria Sultana, M Ashraf Amin, Amin Ahsan Ali, and Akmmahbubur Rahman.

ACL 2026 Main Conference, 64th Annual Meeting of the Association for Computational Linguistics, Long Papers, pp. 43652–43674.

BanHate: An Up-to-Date and Fine-Grained Bangla Hate Speech Dataset 2025

Faisal Hossain Raquib, Akm Moshir Rahman Mazumder, Md Tahmid Hasan Fuad, Md Farhan Ishmam, and Md Fahim.

BLP-2025, Proceedings of the Second Workshop on Bangla Language Processing, ACL, pp. 237–248. [Dataset](#)

LLM4Review: A Pretrained Transformer-Based Framework for Multi-Language Code Review 2025

Mithila Mehedi, Md. Minul Islam Shovon, Faisal Hossain Raquib, Md. Rafiur Rahman

16th ICCNT 2025, 16th International IEEE Conference on Computing Communication and Networking Technologies.

Research Experience

Bangla Hate Speech Detection and Explainability Research

2024 – 2026

- Contributed to the development of BanHate and BanHADEX(first human-annotated hate explainability dataset for the Bangla language), large-scale Bangla hate-speech resources containing 19,203 YouTube comments with binary labels, seven fine-grained hate categories, and seven target and explanation.
- Worked on data collection, preprocessing, annotation, dataset construction, model evaluation, and LLM-based classification and explanation experiments.
- Explored prompting and LoRA-based fine-tuning of open- and closed-source language models for low-resource Bangla NLP.

Undergraduate Thesis Research — Optimizing Sentiment Analysis of Bangla Text: Machine Learning & Deep Learning Perspective May 2024

- Developed Bengali sentiment-analysis pipelines using preprocessing, TF-IDF, Random Forest, SVM, and Logistic Regression.
- Designed a hybrid CNN–BiLSTM deep-learning model to capture local and sequential textual features.

Projects

ChatPDF — AI-Powered PDF Question Answering [\[Link\]](#) [\[Demo\]](#)

- Built a Streamlit RAG application using LangChain, Gemini API, FAISS, and optimized document chunking for multi-PDF question answering.

CodeWizard — AI-Powered Coding Assistant [\[Link\]](#) [\[Demo\]](#)

- Developed a Gradio-based coding assistant using a locally hosted LLM with dynamic prompting and conversation-history management.

Cold Email Generator — LLM-Based Outreach Automation [\[Link\]](#) [\[Demo\]](#)

- Built a Streamlit application using LangChain, Groq LLaMA3, and ChromaDB to generate context-aware emails from job postings.

Freelancer Hiring [\[Link\]](#)

- Developed a PHP-based freelancer platform with profiles, messaging, and administrative content management.

Student Management System [\[Link\]](#)

- Built a database-backed web application with separate student/admin interfaces for notices, routines, results, and student management.

Research Presentation

- **BanHADEX: Towards Explainable HAtE Speech Detection in Bangla Using Human Annotated EXplanation**
ACL 2026, 64th Annual Meeting of the Association for Computational Linguistics, San Diego, California, USA
- **BanHate: An Up-to-Date and Fine-Grained Bangla Hate Speech Dataset**
BLP-2025, Proceedings of the Second Workshop on Bangla Language Processing, ACL

Ongoing Research

- Multi-level hate speech detection in regional Bengali dialects
- BanGrammer: A Benchmark Dataset for Evaluating Large Language Models on Bengali Grammar

Technical Skills

Programming: Python, C, C++, MATLAB

Machine Learning / Deep Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV, pandas, NumPy

NLP / LLM: LangChain, Hugging Face, RAG, FAISS, ChromaDB

Development: Streamlit, Gradio, PHP, JavaScript, HTML, CSS

Tools: Git, LaTeX, Kaggle, Google Colab, Jupyter Notebooks

Leadership & Volunteer Experience

Founder, Prarthona Paribar

2016 – Present

- Founded a non-profit initiative supporting underprivileged communities through food and clothing distribution during floods, Ramadan, and special occasions.

Vice President, Photographic Society of RUET (PSR)

October 2023 – July 2024

- Organized exhibitions, workshops, and creative activities while promoting photography and visual storytelling among students.

Vice President, CESA

September 2023 – August 2024

- Represented engineering students from the Chittagong district and supported student development, collaboration, and leadership activities.

Technical Team Member, Technocracy

May 2019 – July 2019

- Supported a national-level competition as part of a 50+ member volunteer team, contributing to technical and event operations.

Academic References

Hafsa Binte Kibria

Assistant Professor

Department of Electrical and Computer Engineering (ECE), RUET

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[Profile](#)

Md. Fahim

PhD Student

The University of Texas at Dallas (UT Dallas)

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Oishi Jyoti

Assistant Professor

Department of Electrical and Computer Engineering (ECE), RUET

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